



MICROSOFT FABRIC GOVERNANCE

Fabric Atlas

Workspace intelligence and governance for Microsoft Fabric

Catalog | Lineage | Governance | Access | Operations

FABRIC NATIVE

OPEN SOURCE

METADATA ONLY

Product version 1.12.0. Example deployment evidence: FGI-MAIN, captured 6 September 2026.

September 2026

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This paper documents Fabric Atlas version 1.12.0. FGI-MAIN is used as one example deployment, captured on 6 September 2026. Its counts and findings describe that workspace at one point in time. They are not product defaults, limits or benchmarks.

1 Executive summary

Microsoft Fabric puts data engineering, analytics, real-time processing and business intelligence in one service. The workspace is the practical unit where teams build and operate that estate, but the metadata needed to understand it is spread across item pages, admin APIs, run history, permissions and semantic models.

Fabric Atlas brings that metadata into one workspace application. It indexes Fabric items, internal schema objects, lineage, access, sensitivity, recent operations, configuration and team notes. The application runs inside the Fabric portal and stores its validated metadata snapshots in Fabric through Rayfin.

The product is deliberately narrow about data collection. It stores governance metadata, not workspace business data. It does not copy table rows, notebook code, pipeline definitions, Power Query expressions, connection details or report visual bindings. Measure DAX is retained only when Fabric exposes it as semantic model metadata, because Atlas uses those expressions to resolve measure and column dependencies.

The example deployment used for the figures contained 76 Fabric items across 19 item types, 64 item-lineage links and 1,164 indexed assets. The fresh version 1.12.0 deployment had one validated baseline snapshot. FGI-MAIN is included to make the workflows concrete, not to define how an Atlas workspace must look. Another supported Fabric workspace will have its own item mix, metadata coverage and governance history.

1.1 What Atlas provides

Area	Practical outcome
Workspace inventory	One searchable list of Fabric items with type, ownership, health, labels, tags and context
Asset inventory	Relational, semantic, KQL, ontology, graph and data-agent objects grouped under their parent item
Lineage	Source-to-consumer paths across pipelines, notebooks, stores, endpoints, semantic models and reports
DAX dependencies	Verified measure-to-column and measure-to-measure references when names resolve uniquely
Governance	Findings, new-risk radar, snapshot comparison, metadata coverage and posture targets
Access review	Effective access that combines workspace inheritance and direct item grants
Operations	Recent refresh, pipeline and notebook activity with status, duration and error context
Collaboration	Shared, append-only notes attached to the workspace or a Fabric item

2 Why a workspace atlas is useful

Fabric already exposes rich metadata. The problem is not that the facts do not exist. The problem is that a reviewer has to move between several experiences to answer a basic question.

Consider a semantic model change. A reviewer may need to identify its upstream Lakehouse, the notebooks that write to that Lakehouse, the reports bound to the model, the measures that depend on a changed

column, the people who can edit the model and the most recent refresh result. Each fact can be found, but rarely in one place and rarely with the same point in time.

Atlas treats the workspace as a connected metadata graph. The item catalog is one view of that graph. Lineage, access, findings and run history are other views over the same validated snapshot. This avoids a common review failure where two portal pages describe different moments in the workspace lifecycle.

The snapshot model also changes the questions a team can ask. Instead of only asking “what is here now?”, a team can ask:

- Which items, grants, schema objects or lineage edges changed since the last accepted snapshot?
- Did a change create a new high-priority risk?
- Is a low score caused by a real gap, or did the API not expose the metadata?
- Which reports are downstream of a model?
- Which measures use a selected model column?
- Is access inherited from the workspace, granted directly on the item, or produced by both paths?

Atlas does not replace Fabric administration or deployment tooling. It is a workspace reading and review layer that keeps evidence close to the people who operate the assets.

3 Example deployment: FGI-MAIN

The screenshots in this paper use FGI-MAIN as a practical case study. It is a deployed Fabric App connected to one real workspace and had been synchronized about nine hours before the capture. It is not a reference architecture or a required workspace shape.

Observed value	Example count
Fabric items	76
Item types	19
Item-lineage links	64
Indexed assets	1,164
Tables	144
Columns	376
Measures and KPIs	62
Validated snapshots	1
Unique principals in Access Review	2
Reachable principal and item pairs	78
Recent job runs	25
Failed recent job runs	3

The Overview page is designed for orientation rather than exhaustive detail. It combines workspace identity, sync freshness, posture targets, priority signals, metadata coverage and activity. Links on the page open the matching filtered view, so the summary remains connected to evidence.

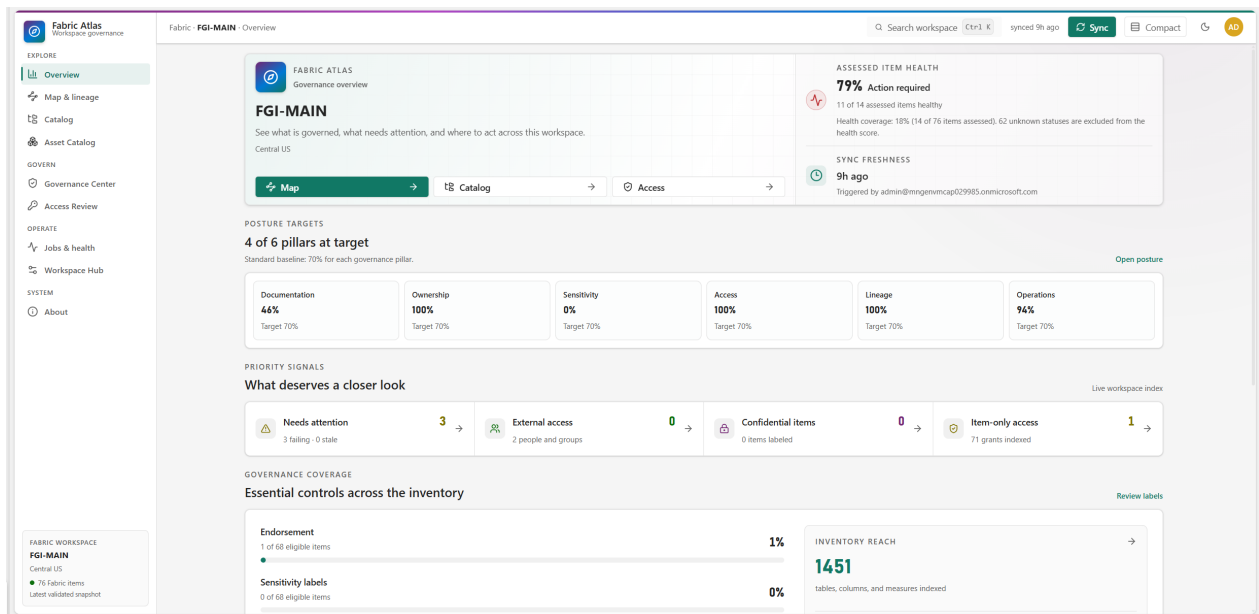


Figure 1: Overview from the FGI-MAIN example deployment.

The assessed-item health score in Figure 1 is 79 percent. Eleven of the 14 items with health evidence are healthy; the other 62 workspace items remain unknown and are excluded from the score. Atlas does not turn missing evidence into a failure. Missing evidence and negative evidence are different states.

The six posture pillars are evaluated independently. In this snapshot, ownership, access, lineage and operations met their targets. Documentation and sensitivity did not. That distinction gives the workspace team a concrete starting point without blending unrelated controls into one opaque score.

4 Catalog and internal asset inventory

4.1 Workspace catalog

The Catalog lists every item returned by the Fabric Items API, including item types that do not yet have a dedicated deep scanner. Unknown or newer types stay visible with a neutral type treatment instead of disappearing.

Items can be filtered by type, searched by name or tag and opened in a detail drawer. The drawer keeps ownership, description, lineage, effective access, configuration and recent jobs next to the selected item. Groups remain collapsed until the user searches or chooses to expand them.

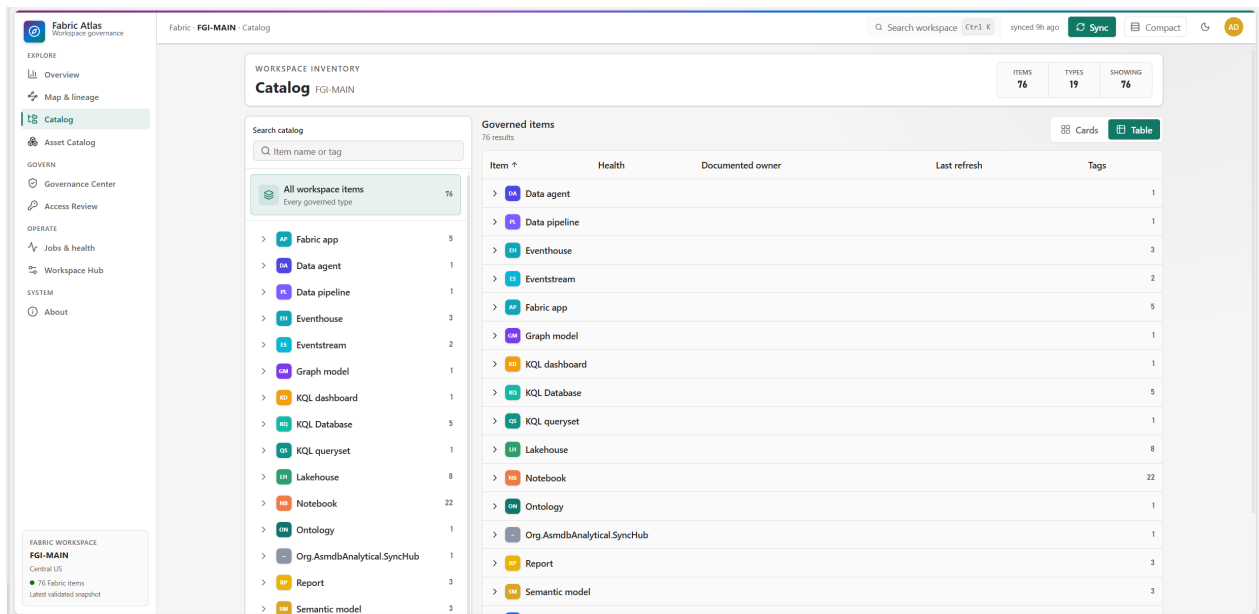


Figure 2: The example catalog contains 76 items across 19 Fabric item types.

This view answers “what exists?” without forcing every object into the same shape. A Notebook, Semantic Model, Lakehouse and Fabric App remain distinct types, but they can still participate in common workflows such as search, lineage inspection, access review and change comparison.

4.2 Asset Catalog

The Asset Catalog moves below the item boundary. It groups synchronized relational, semantic, KQL, ontology, graph and data-agent objects under their parent Fabric item. In the captured workspace, 39 items exposed at least one asset. The inventory contained 1,164 assets, including 144 tables, 376 columns and 62 measures or KPIs.

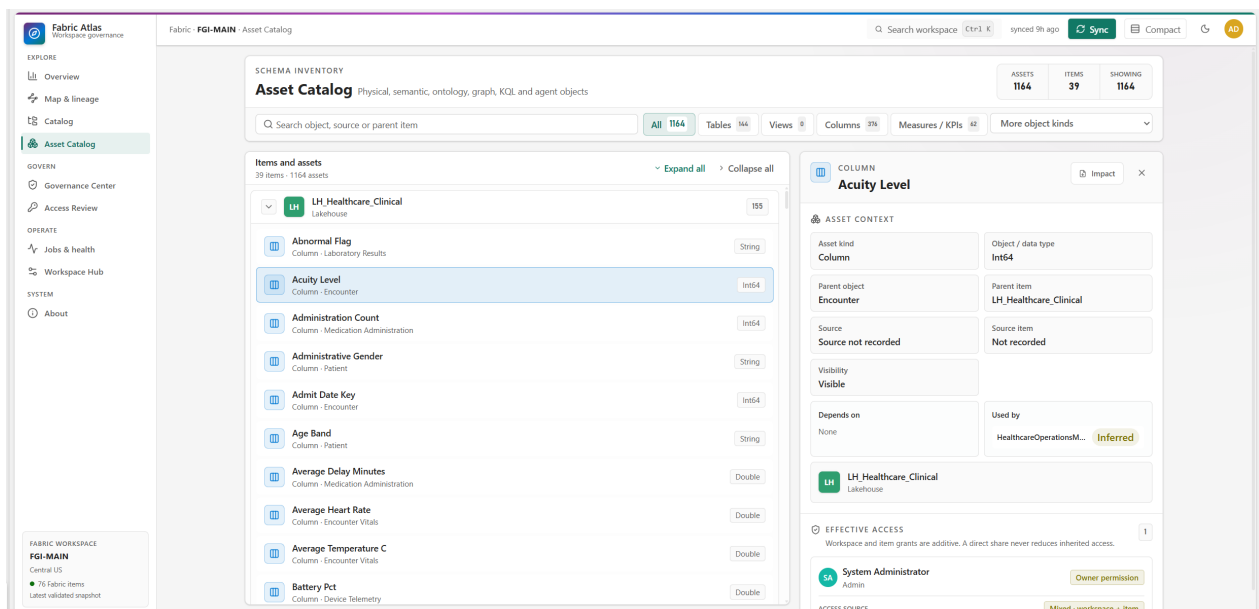


Figure 3: A Lakehouse column selected in the example Asset Catalog.

Atlas records the provenance of an object where possible. A Lakehouse table may come from the Fabric Tables REST API. A semantic model column or measure may come from scanner metadata. Warehouse and SQL Database objects may come from the scanner, with a labelled downstream model subset when complete SQL catalog access is not available.

The object detail keeps four kinds of evidence together:

1. Object identity, parent table and parent item.
2. Data type, visibility, description and collection source when exposed.
3. DAX dependency evidence for semantic model columns and measures.
4. Effective access to the parent Fabric item.

4.3 Measure metadata

Measure expressions are useful for more than documentation. Atlas parses the DAX expression after removing comments and string literals, then resolves qualified columns and measures against the synchronized schema. A dependency is emitted only when the reference matches one real object unambiguously.

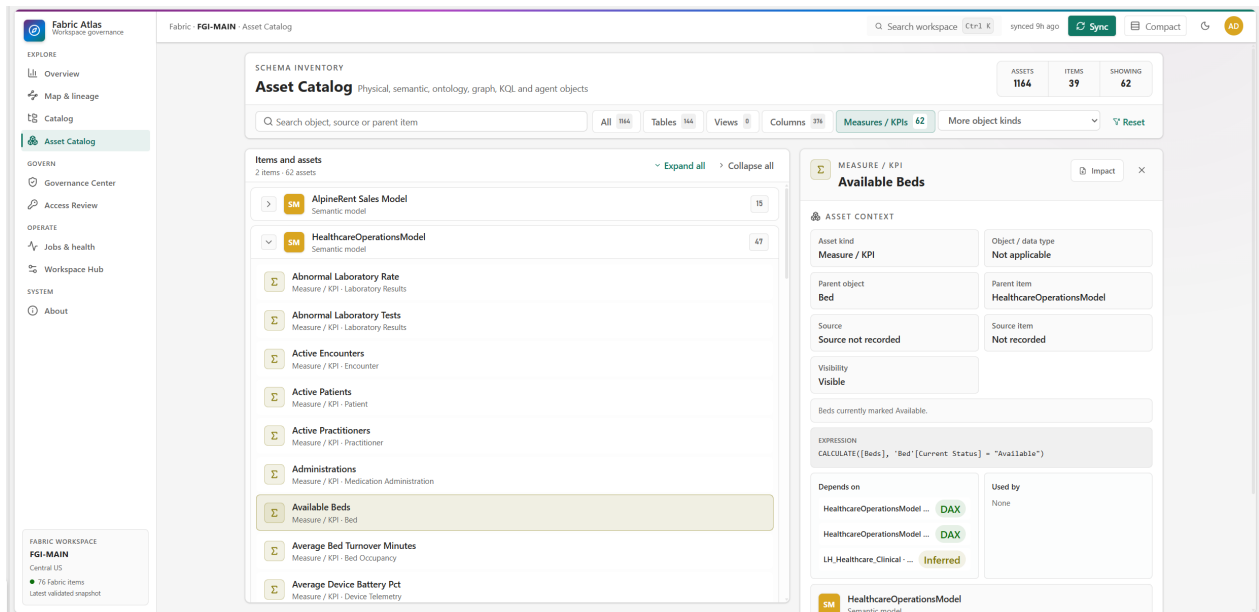


Figure 4: The Available Beds measure with its DAX expression and resolved dependencies.

Figure 4 shows the distinction between a DAX edge and an inferred source hop. The **Available Beds** measure filters the **Bed[Current Status]** column and reuses the **Beds** measure. Those same-model references receive verified DAX labels. Atlas also connects the semantic object to the matching upstream Lakehouse object when item lineage and the schema match are unambiguous. That cross-item hop is labelled **inferred**. Ambiguous references are discarded. Atlas does not guess which object the author meant, and it does not create object lineage from name similarity alone.

5 Lineage and impact analysis

5.1 Item lineage

The item graph normalizes relationships from source to consumer and arranges assets by lifecycle stage:

1. Orchestrate
2. Transform
3. Store
4. Endpoint
5. Model
6. Consume

Selecting an item highlights its upstream and downstream paths without moving the graph. The inspector shows the same item context used elsewhere in the application, plus impact counts and textual relationship summaries.

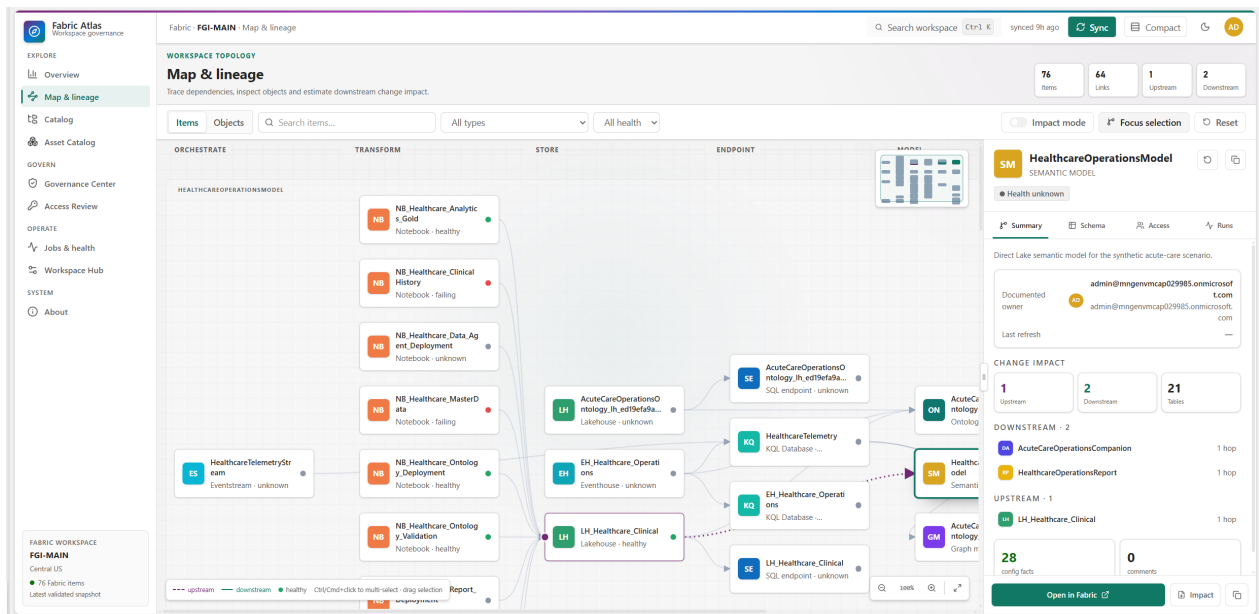


Figure 5: The HealthcareOperationsModel selected in item lineage.

In the captured graph, HealthcareOperationsModel has one direct upstream Lakehouse and two direct consumers: a report and a data agent. Its complete impact path contains ten upstream items, two downstream items and fourteen relationships. Atlas traverses that reachable graph through indexed adjacency maps, so selection does not require a full scan of every edge.

5.2 Object lineage

Object mode expands synchronized tables, columns and measures while retaining the surrounding item graph. Purple dashed paths indicate upstream movement. Teal paths indicate downstream movement. The graph also exposes relationship text, so direction is not encoded by colour alone.

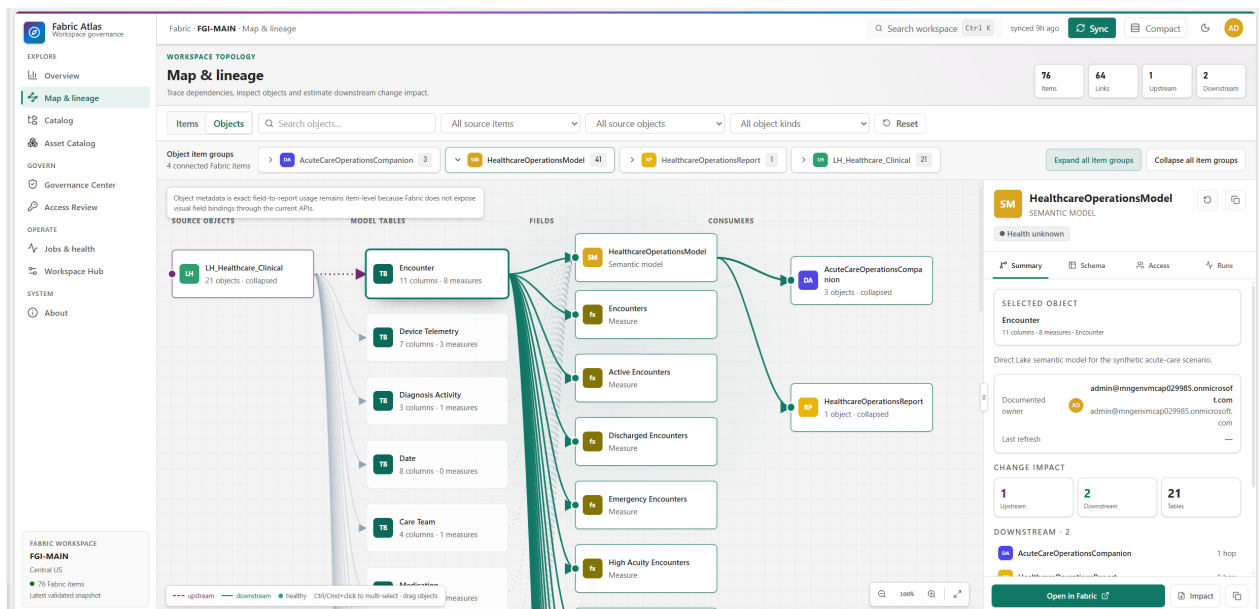


Figure 6: Object mode connects Lakehouse tables, semantic model fields and downstream reports.

The graph is precise about its boundary. Object metadata can be exact while field-to-report usage remains item-level. The Fabric APIs used by this release do not expose complete visual field bindings. Atlas can prove that a report is bound to a semantic model, but it does not claim that a particular visual uses a selected measure.

5.3 Downstream impact at two levels

Atlas supports two related forms of impact analysis.

At item level, selecting a semantic model returns all reachable downstream items. In the FGI-MAIN example, the HealthcareOperationsModel impact report lists HealthcareOperationsReport and AcuteCareOperationsCompanion as one-hop consumers. It also lists ten upstream dependencies and the relationship evidence used to build the result.

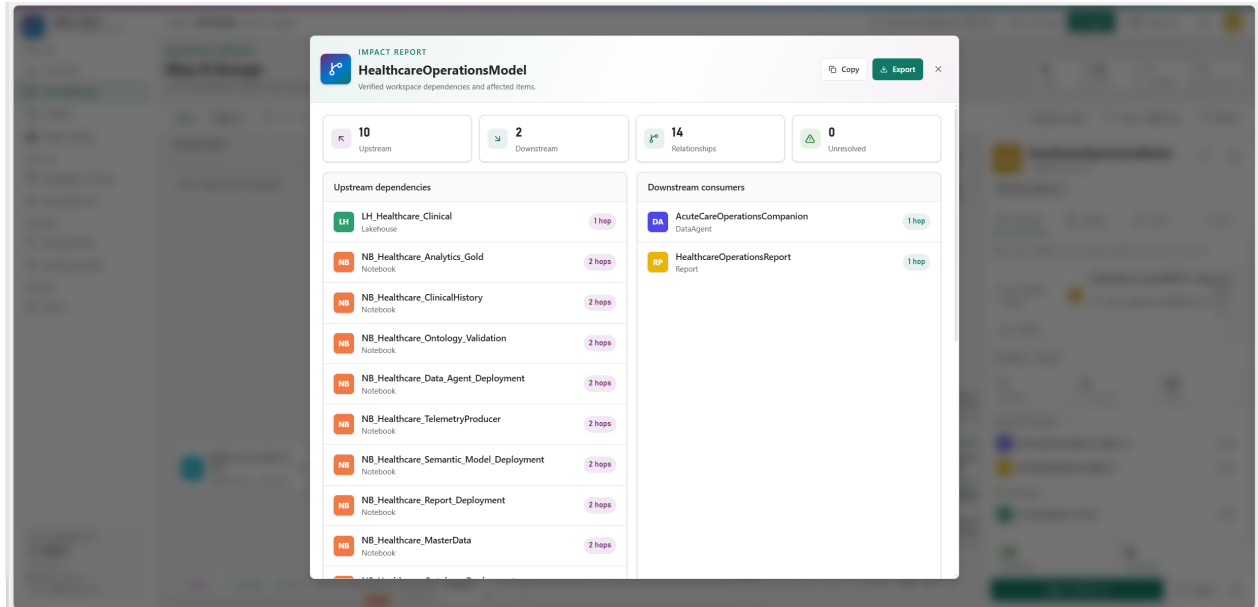


Figure 7: Item-level impact for HealthcareOperationsModel.

At object level, Atlas builds a reverse dependency index from resolved DAX references. Selecting a model column can therefore show which measures use it. The captured Bed[Current Status] model column is used by three measures, including Available Beds, and the selected object is also exposed to the AcuteCareOperationsCompanion data agent.

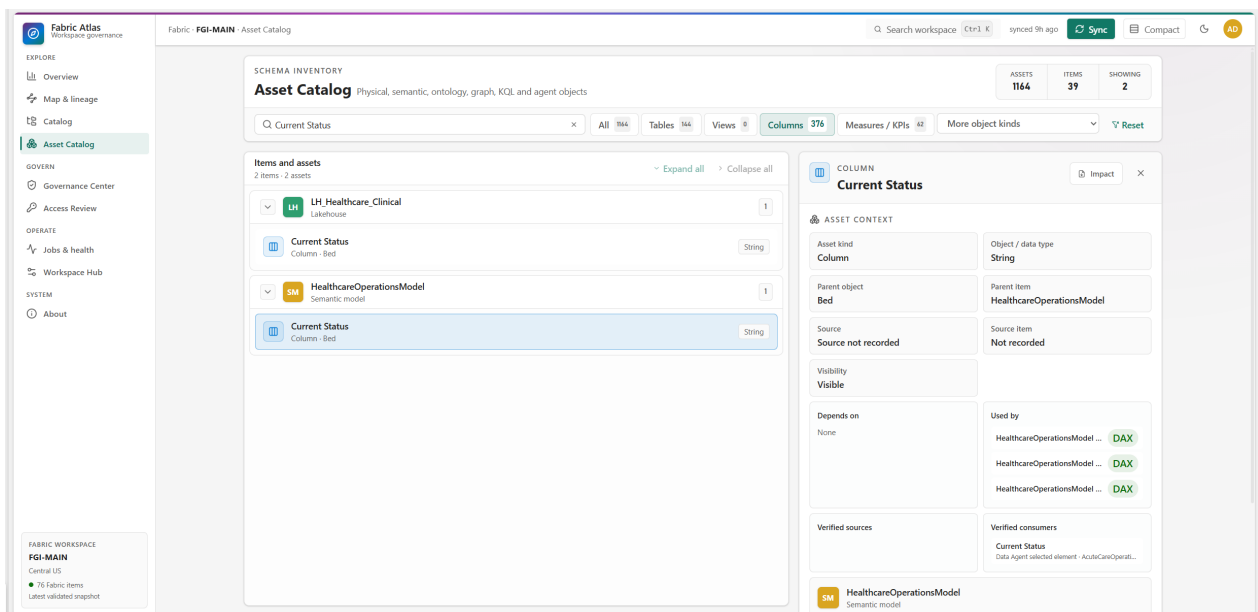


Figure 8: A model column with a verified DAX consumer.

The exportable impact report keeps the relationship type beside the object evidence. Same-model DAX dependencies appear in the "Used by" list, while the report in Figure 9 shows the verified data-agent selected-

element relationship.

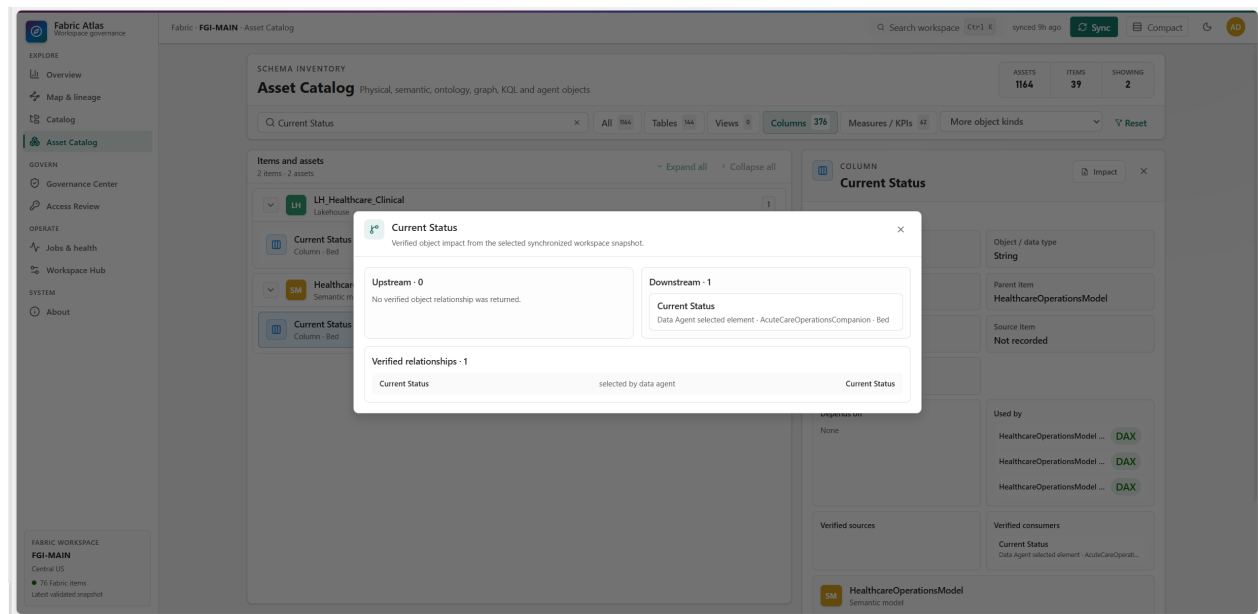


Figure 9: Exportable object impact report with a verified DAX consumer.

This distinction prevents an overclaim. Atlas can identify reports that depend on a semantic model and measures that depend on a column or another measure. It cannot currently identify the exact report visual that uses that field.

6 Governance Center

The Governance Center uses the same synchronized graph to support review over time. It separates five jobs that are often mixed together in governance dashboards: current findings, change comparison, history, collection coverage and posture targets.

6.1 Governance Radar and findings

Governance Radar compares the latest adjacent validated snapshots. It reports new high or critical findings and risky changes involving access, sensitivity, lineage or removals. Existing findings do not reappear as new every time a user opens the page.

The captured deployment contains one validated baseline snapshot, so Radar has no adjacent pair to compare yet. The Findings section contains 76 current checks, including failed-operation evidence. The next successful sync will produce the first before-and-after risk delta.

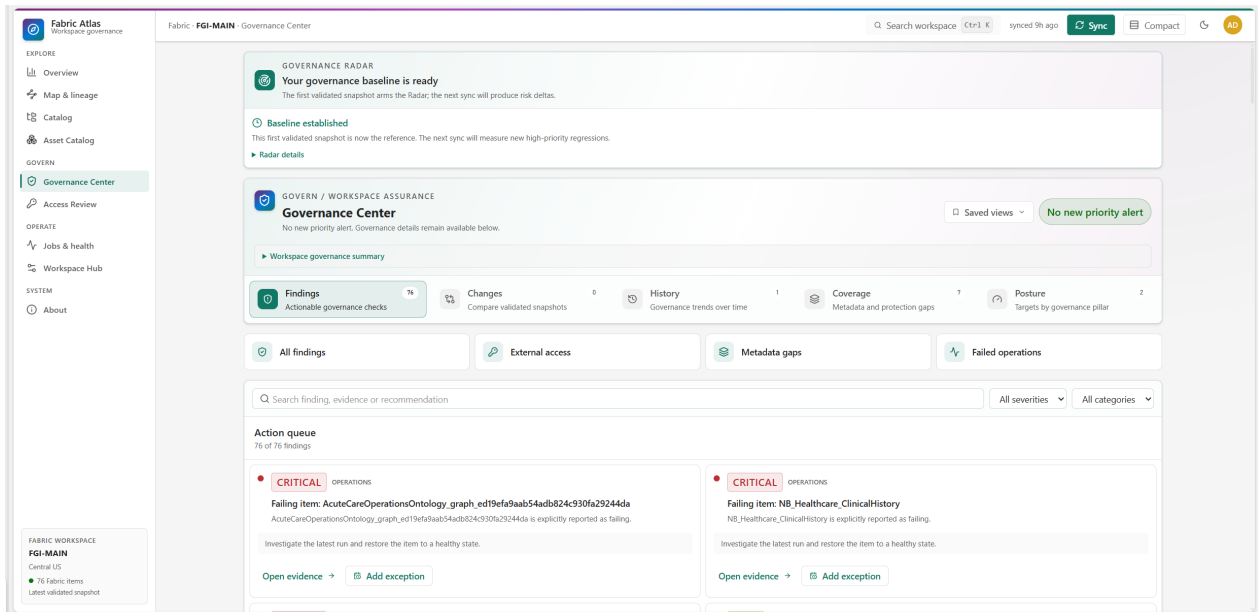


Figure 10: Governance Radar and the current findings queue.

Radar acknowledgements and mutes are personal. One user can acknowledge a finding occurrence without changing another user's view or hiding the underlying finding from the workspace.

6.2 Change Center

Change Center compares any two validated snapshots. It covers item metadata, schema objects, access grants, sensitivity, lineage and jobs. The selected snapshot pair and filters are kept in the URL so a reviewer can share the exact comparison.

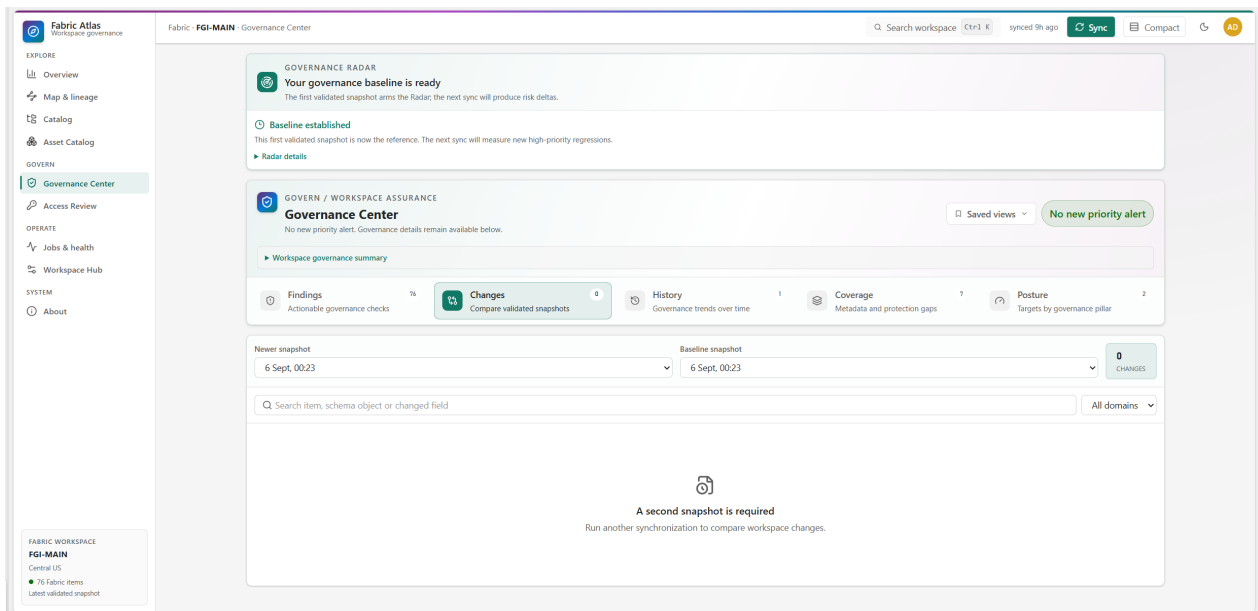


Figure 11: Change Center waiting for a second validated snapshot.

Older detailed catalogs are loaded only when selected. Workspace manifests carry compact trend summaries, which keeps the history ledger responsive without hydrating every historical row during application startup.

6.3 Metadata coverage

Coverage reports what the sync contract collected and what the APIs exposed. It does not convert unsupported metadata into a governance defect.

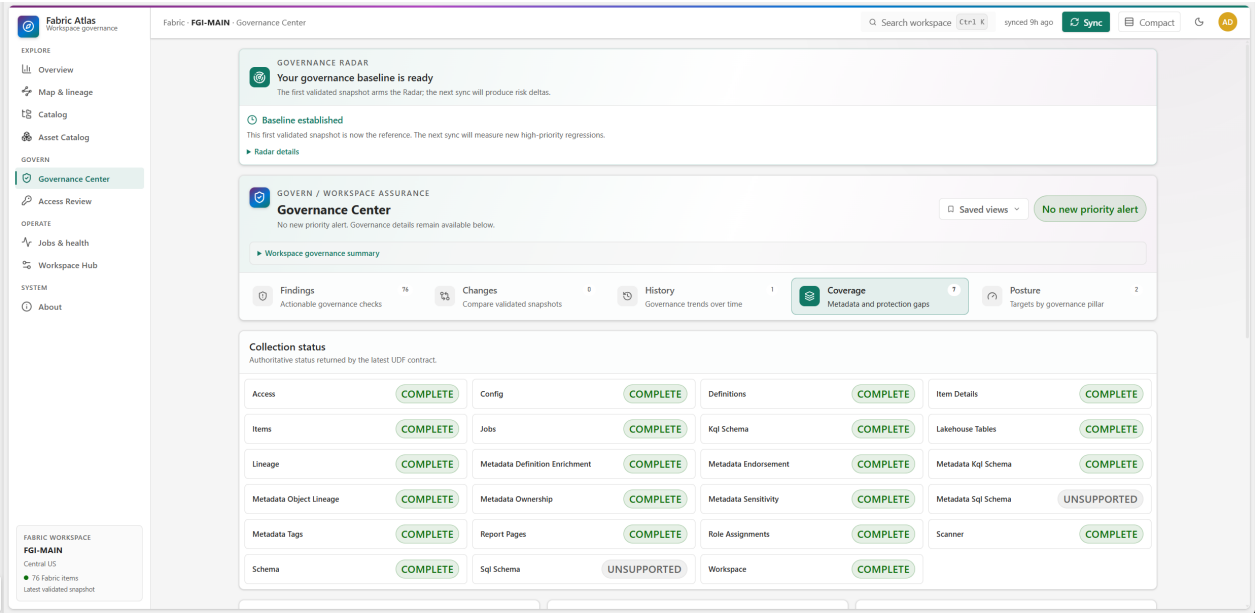


Figure 12: Collection status and coverage for the example workspace.

The captured sync completed the required contract and most optional metadata families. Direct SQL schema collection is marked unsupported in this snapshot, while scanner schema, item details, definitions, KQL schema, report pages and object lineage are complete. A complete collection state still does not mean that every item has a value. Sensitivity metadata was collected, but none of the 68 eligible items had a label. This split between collection status and value coverage is important. It lets a reviewer distinguish “the scanner did not provide this metadata” from “the scanner provided the field and the item has no value.”

6.4 Posture targets

Atlas scores six reproducible pillars: documentation, ownership, sensitivity, access, lineage and operations. Non-applicable evidence is excluded rather than counted as zero.

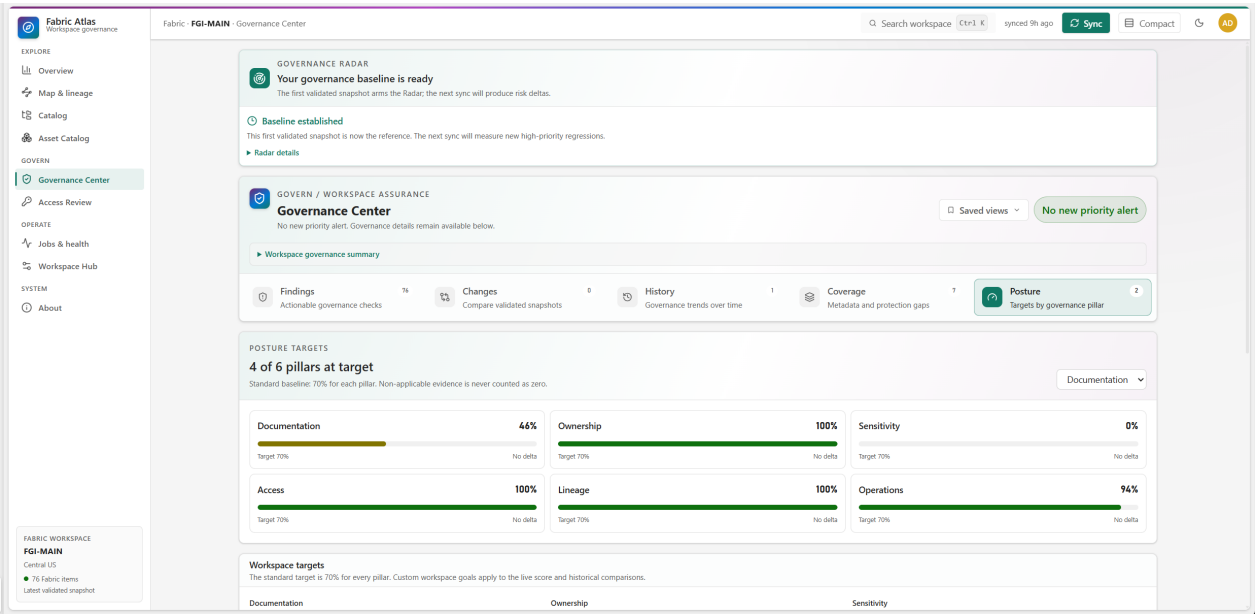


Figure 13: Four of six governance posture pillars met their targets.

Each pillar has a target and a drill-down path. A score is useful only when the team can open the items behind it, so Atlas keeps the score connected to Catalog, Access Review, Jobs or the matching governance evidence.

7 Access review

Fabric permissions are additive. A workspace role may grant access to every item, while an item-level share adds another path. A direct grant must never reduce inherited access. Atlas calculates the highest effective permission for every reachable principal and item pair. It keeps the contributing grants and their origins, then exposes filters for access level, source and risk flags.

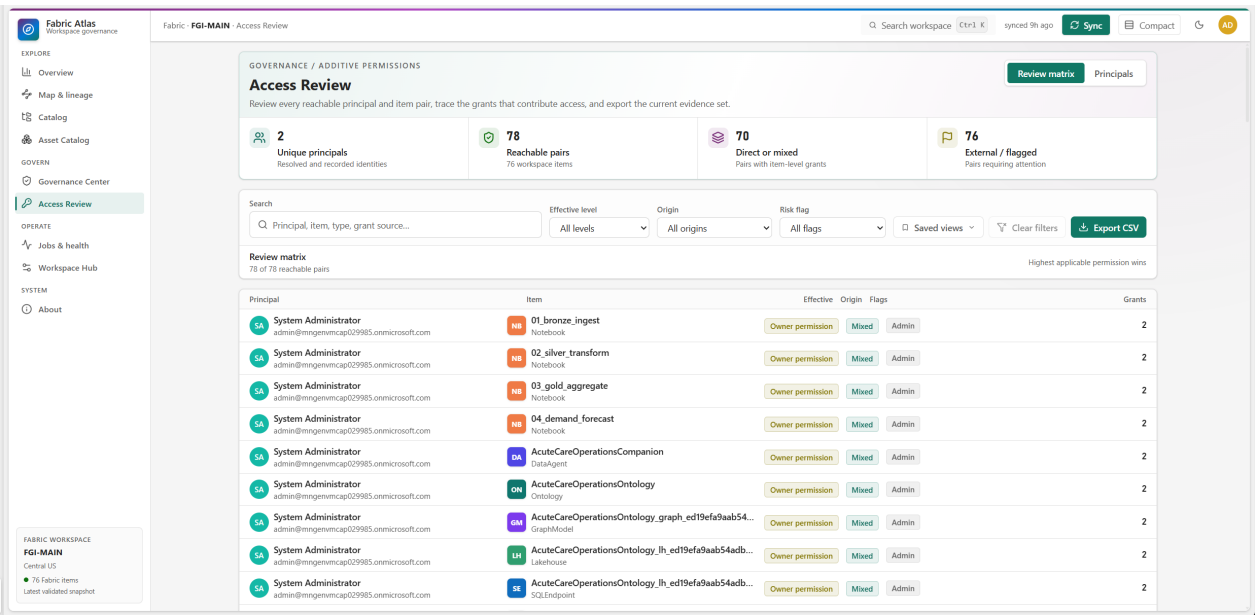


Figure 14: Access Review in the example deployment, with effective level, origin and grant count.

The captured workspace contained two resolved principals and 78 reachable principal and item pairs. Seventy pairs had direct or mixed item-level evidence. The filtered matrix can be exported as CSV.

Review decisions are personal. A user can mark a pair as Reviewed, Accepted or Needs action and add a note. Rayfin row policies bind those records to the authenticated subject, so one user's decision is not presented as a team-wide approval.

7.1 Principal-centred review

The Principals mode groups the filtered access pairs by identity. Each group shows the number of reachable items and the strongest effective access level. Opening a group reveals the individual items, access origin and contributing grant count.

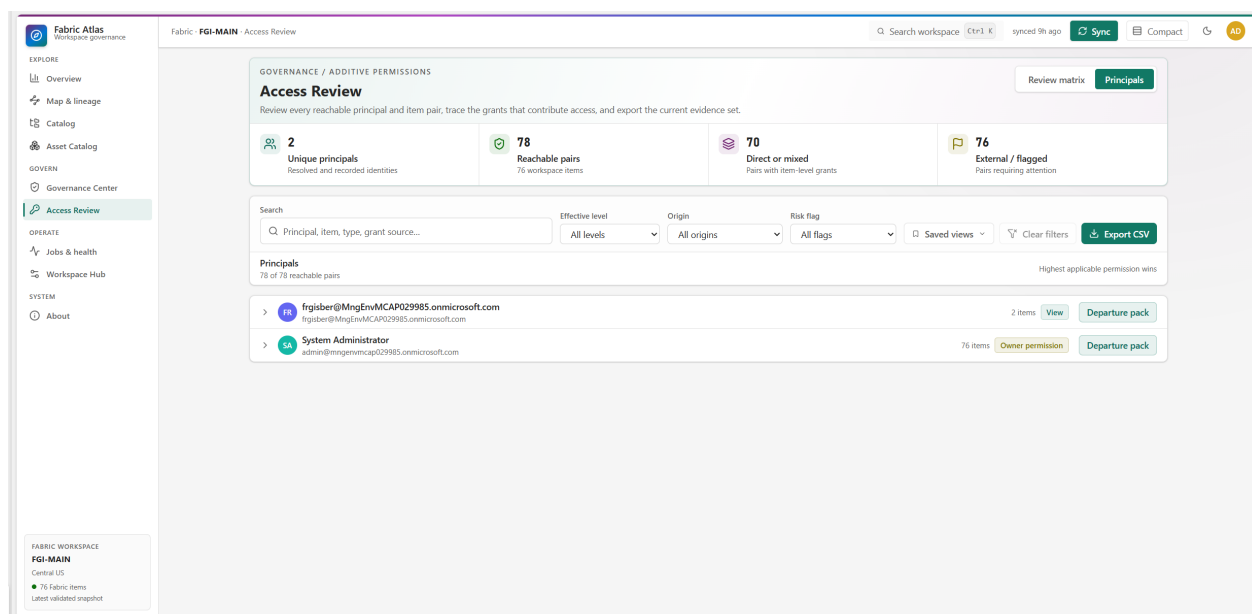


Figure 15: Access Review grouped by principal in the example deployment.

The example contains two principals. One reaches two items with View access. The System Administrator reaches all 76 items with Owner as its strongest effective level. These are access results. Atlas does not treat the item count or strongest access badge as proof of documented item ownership.

Each user or guest group has a Departure pack action. Other principal types use the label Removal impact because group membership is not expanded and the report evaluates the principal itself.

7.2 Departure packs

A Departure pack is calculated from the active validated snapshot. For a person, documented ownership requires the type-specific owner email exposed by Fabric. A matching display name alone is not ownership evidence.

The report combines four result sets:

Result	What it means
Owned or effective owner	Items supported by documented person ownership or effective Owner access for a non-person principal
Sole owned	Owned items with no other resolved owner and no uncertain owner reference
Urgent risks	Sole-owned items that have at least one downstream consumer
Blast radius	Unique downstream items reachable from the ownership roots

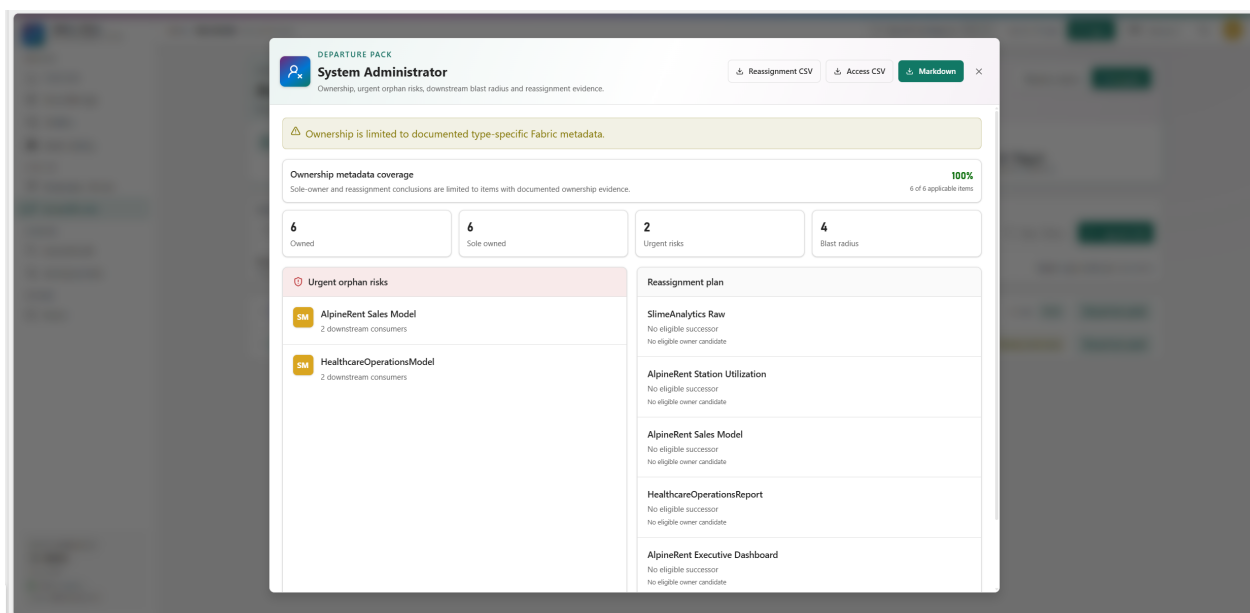


Figure 16: Departure pack for the System Administrator in the example deployment.

In this example, ownership metadata is available for all six applicable items. The pack reports six owned and sole-owned items, two urgent orphan risks and a blast radius of four. The urgent roots are the AlpineRent and Healthcare semantic models, each with two downstream consumers. The reassignment plan states that no eligible successor was found. Atlas keeps that result explicit instead of inventing a candidate.

When eligible owners exist, the reassignment logic first checks the nearest upstream owners. If that produces no candidate, it considers the most frequent eligible owner in the connected lineage component. If neither route produces a resolved internal user, the result remains “No eligible successor.”

The dialog exports a reassignment CSV, the principal’s effective-access CSV and a complete Markdown departure pack. If the selected principal is unresolved or ambiguous, ownership and reassignment conclusions are blocked. The access evidence can still be exported.

8 Operations and workspace context

8.1 Jobs and health

Jobs and health groups recent refresh, pipeline and notebook activity. Users can search by item or job type, filter by status and save personal views.

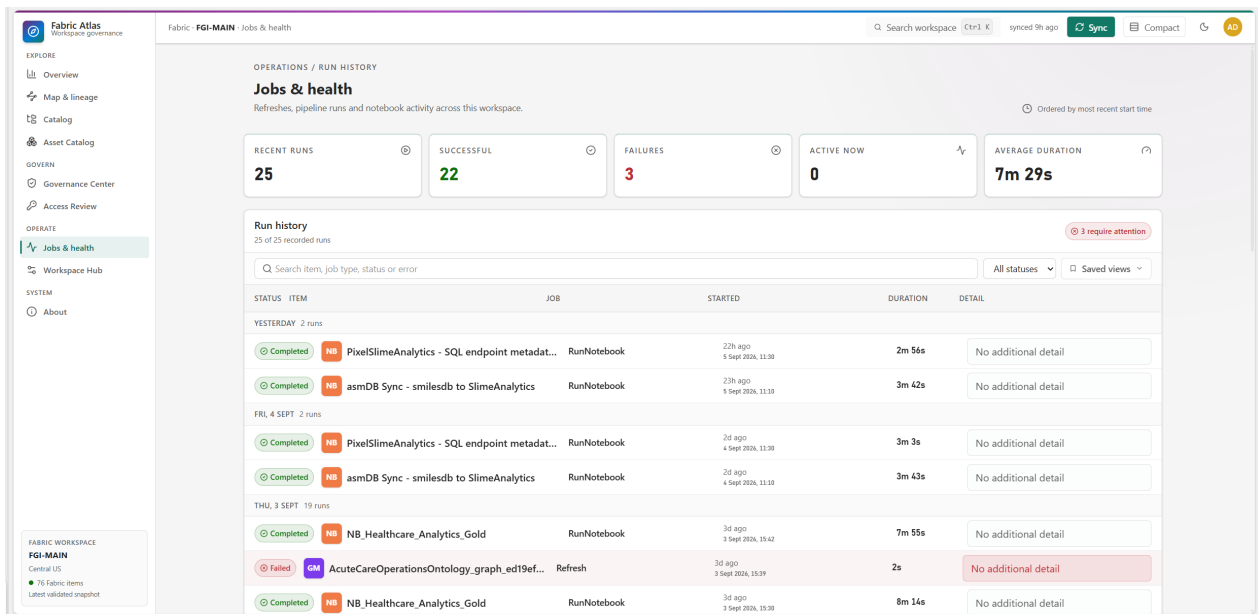


Figure 17: Twenty-five recent runs with success, failure and duration evidence.

The example deployment recorded 25 recent runs: 22 completed and three failed, with an average duration of 7 minutes and 29 seconds. Atlas retains the error message when an endpoint returns one. An optional job endpoint failure does not invalidate an otherwise authoritative metadata snapshot, but the missing collection state remains visible.

8.2 Workspace Hub configuration

Workspace Hub keeps technical configuration and human context in the same place. Configuration facts are grouped by item and section rather than shown as one unstructured JSON payload.

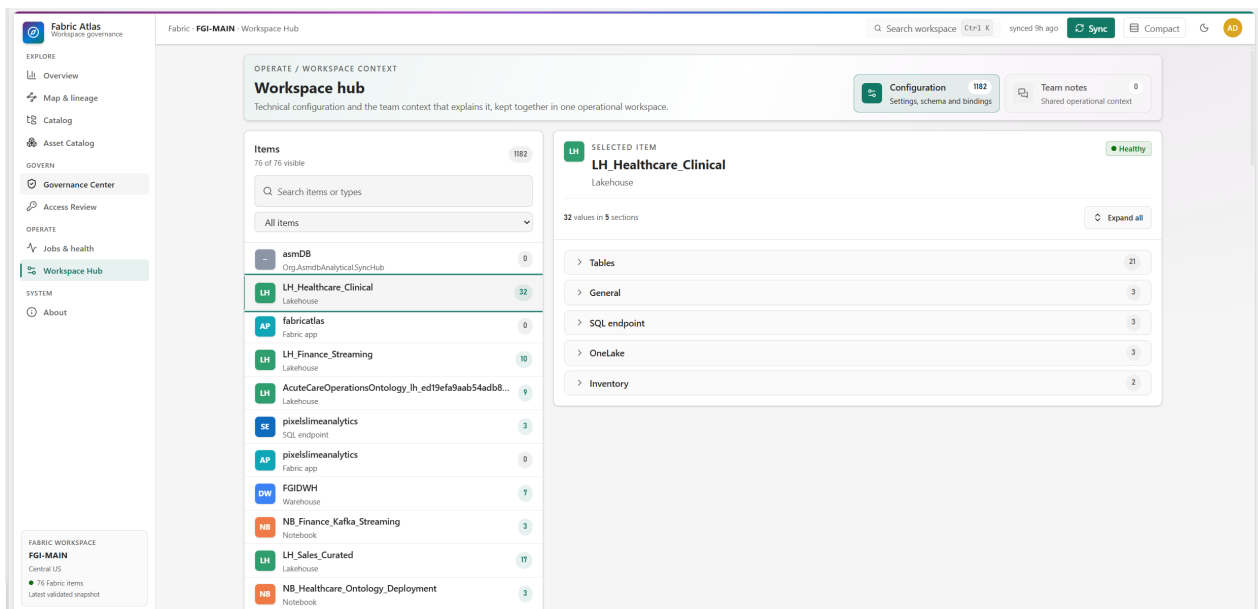


Figure 18: Lakehouse configuration grouped into Tables, SQL endpoint, Inventory, General and OneLake sections.

The captured workspace contained 1,182 configuration facts. The selected Lakehouse exposed 32 values in five sections. Schema lists are chunked when necessary so complete metadata can fit within bounded SQL text fields.

8.3 Team notes

Team notes attach operational context to the workspace or a selected Fabric item. They are stored in the Fabric-backed Rayfin database and survive metadata refreshes.

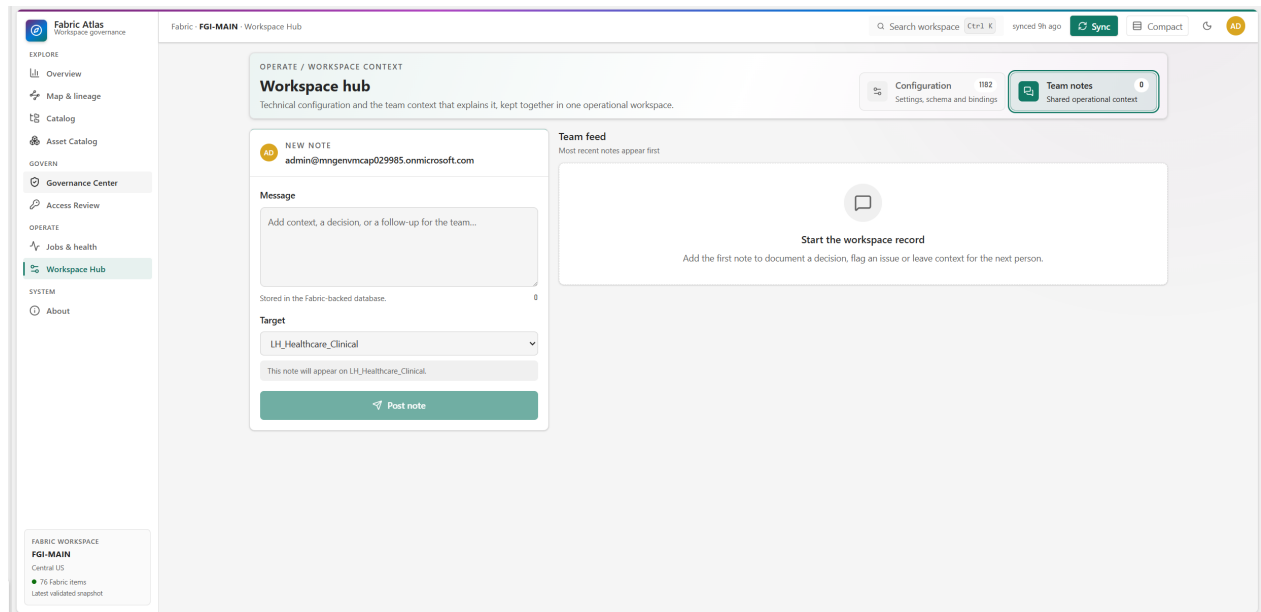


Figure 19: Shared team notes in the Workspace Hub of the example deployment.

The fresh deployment had no team notes at capture time, so Figure 19 shows the empty feed and the workspace-level note composer. Notes are append-only in version 1.x. The stored author label comes from the authenticated email. Update and delete operations are not exposed because Atlas does not yet have a reviewed editing policy and user experience for shared notes.

9 Architecture and synchronization

Fabric Atlas is a React and Vite single-page application deployed as a Rayfin Data App. Fabric brokered authentication supplies the signed-in Entra identity. A published Fabric User Data Function performs the metadata scan because a browser application cannot call all required Fabric management APIs directly.

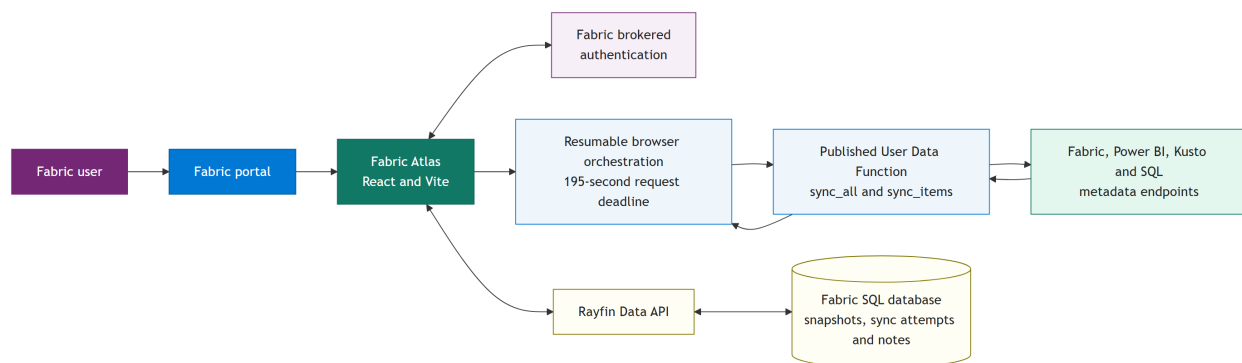


Figure 20: Deployment and metadata flow

9.1 Synchronization contract

The User Data Function returns a versioned contract with required sections, optional enrichment and metadata capability status. The browser first requests an authoritative workspace topology, then sends type-grouped `sync_items` slices for deeper metadata. A required-section failure or any failed item enrichment rejects publication.

Each UDF invocation has a 180-second monotonic budget, leaving 20 seconds below Fabric's 200-second platform timeout. A global budget expiry, one slow request and a Retry-After window that does not fit in the remaining slice are separate conditions. All three return a continuation instead of pretending the item is complete. The browser uses a 195-second deadline, renews expiring tokens between slices and supports cancellation before the manifest is published.

The UDF URL is validated before token acquisition. It must be an HTTPS Fabric User Data Function endpoint for the configured workspace, and redirects are rejected in both browser and server transports. The main Fabric token is read-oriented. A separate optional token is requested only for definition metadata that requires `Item.ReadWrite.All`; Kusto and SQL tokens remain optional. Upstream and final payloads are capped at 25 MiB, and the browser independently caps the streamed response before JSON parsing.

9.2 Immutable snapshots

Each successful refresh writes a new immutable snapshot. Child rows are written first. The Workspace manifest is written last, after row counts and required metadata have been validated. Only that final manifest makes the snapshot visible.

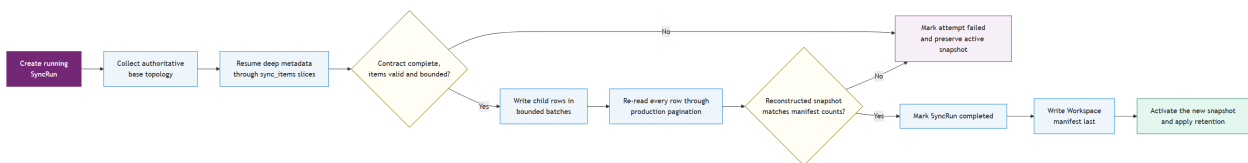


Figure 21: Immutable snapshot lifecycle

If collection, cancellation, persistence or reconstruction fails, the child rows have no visible manifest. Hydration ignores them and loads the newest complete trusted snapshot. Writes use batches of eight requests, and every persisted entity is read back through the same cursor pagination path used at startup before the visibility marker is created.

The SyncRun row exists before Fabric collection begins. It records the correlation ID, start time, terminal state, duration and a bounded failure summary. Status updates are retried. Abandoned snapshot rows are eligible for cleanup only after a grace period, while the attempt record remains available for audit.

9.3 Retention and writer trust

Snapshot retention defaults to 12 and can be configured between 2 and 50. Cleanup starts only after the new manifest is published. Child rows are deleted before their stale manifest, and a cleanup failure does not convert a successful sync into a failed one.

The configured synchronization account is the trusted writer. Publication policies compare the authenticated immutable subject with deployment configuration. Email remains the visible contact and historical writer label, but it is not the authorization key. Former synchronizers can be listed explicitly for historical reads and cleanup during a controlled rotation; they cannot publish a new snapshot.

9.4 Persisted entities

Entity	Stored purpose
Workspace	Complete snapshot manifest and compact governance summary
FabricItem	One row per Fabric item
LineageEdge	Directed source-to-consumer dependency
Principal	User, group, service principal or guest identity
AccessGrant	Workspace or item access evidence
JobRun	Recent refresh, pipeline or notebook activity
ConfigEntry	Bounded configuration facts and schema chunks

Entity	Stored purpose
Comment	Shared workspace or item note
SyncRun	Durable running, completed or failed synchronization attempt
SavedView	Personal navigation and filter preset
AccessReview	Personal review decision for an effective access pair
AccessReviewEvent	Personal append-only access review history
FindingAck	Personal Radar acknowledgement or mute
GovernancePolicy	Shared posture targets for the workspace
GovernanceException	Shared, time-bounded exception with reason and scope

10 Security and data boundaries

Fabric Atlas should be understood as a metadata system. Its security boundary is defined by what it collects, who can read that collection and who can write each kind of state.

10.1 Data that Atlas stores

Atlas stores item identity, documented ownership fields, descriptions, labels, tags, object schema, measure expressions, lineage, principals, grants, recent job metadata, configuration facts, snapshot summaries and team notes.

10.2 Data that Atlas excludes

Excluded content	Reason
Table and event rows	Business data is outside the product purpose
Data source and connection details	The catalog does not need credentials or connection payloads
Power Query and source expressions	Source logic is not required for the governance graph
Notebook source and cells	Atlas inventories the item without copying its code
Pipeline definitions and expressions	The product records the item and exposed lineage, not orchestration source
User Data Function source	Functions remain visible as items without copying implementation code
Complete report visual and field bindings	The current metadata flow does not expose them reliably
KQL rows, query text and policy bodies	Atlas reads schema metadata through Kusto but does not copy business events or executable query content

The scanner response passes through an explicit allowlist before persistence. Unexpected fields are not serialized by default.

10.3 Authentication and authorization

Fabric brokered authentication runs the application under the signed-in Entra identity. The token used for live synchronization must match the current Fabric user and tenant. Sync tokens use session storage so a later user cannot silently inherit the first account from a persistent browser cache.

Snapshot publication is bound to the configured Entra subject, which is stable across email changes. Team-note author IDs are also bound to the authenticated subject, while the displayed author label comes from the authenticated email. External identity is stored as true, false or unknown so absent evidence is not presented as proof that a principal is internal.

State	Read scope	Write scope
Synchronized catalog and history	Every authenticated user admitted to the deployed app	Configured synchronizer only
Shared team notes	Every authenticated app user	Authenticated author, with email and subject claim binding
Saved views	Current user only	Current user only
Access review decisions	Current user only	Current user only
Radar acknowledgements and mutes	Current user only	Current user only

The shared catalog scope is intentional in the single-workspace version. Every user who can open the deployed Fabric App can read the complete synchronized governance graph and shared notes for that workspace. Deployment owners should treat the Fabric App audience as the catalog read boundary.

10.4 Failure behaviour

Atlas favours explicit failure over success-shaped fallbacks:

- A required sync section failure rejects the refresh.
- A failed item enrichment rejects the complete candidate snapshot.
- Malformed authoritative lineage or workspace identifiers fail closed.
- Payloads above the configured bounds are rejected before persistence.
- An incomplete snapshot never becomes active.
- Cancellation before manifest publication leaves the current snapshot active.
- Unknown sensitivity rankings do not create downgrade alerts.
- Ambiguous DAX references and ambiguous principals are omitted rather than guessed.
- Personalization failures do not hide the underlying governance finding.

11 Deployment and operating model

11.1 Prerequisites

A live deployment requires:

1. A Microsoft Fabric workspace on supported capacity.
2. Fabric Apps enabled for the relevant tenant users.
3. A Rayfin deployment with Fabric authentication, data, storage and static hosting.
4. An Entra single-page application registration for delegated sync access.
5. A Fabric Administrator for the metadata scanner, plus the documented tenant settings and Entra consent.
6. The `atlas_sync_functions` User Data Function published in the workspace.
7. One configured synchronization account.

The application can run locally with the bundled preview data, but live synchronization and brokered identity require the Fabric portal.

11.2 Deployment flow

The standard deployment commands are:

```
npx rayfin login --tenant <tenant-id> --select
npx rayfin up --workspace "<workspace-name>"
```

The public runtime values are supplied through the git-ignored Rayfin environment file:

```
RAYFIN_PUBLIC_ATLAS_SPA_CLIENT_ID=<entra-client-id>
RAYFIN_PUBLIC_ATLAS_UDF_URL=https://<host>/functions/sync_all/invoke
RAYFIN_PUBLIC_ATLAS_WORKSPACE_NAME=<workspace-display-name>
RAYFIN_PUBLIC_ATLAS_SYNC_ADMIN_EMAIL=<authorized-sync-user>
RAYFIN_PUBLIC_ATLAS_SYNC_ADMIN_SUBJECT=<entra-object-id>
RAYFIN_PUBLIC_ATLAS_SNAPSHOT_RETENTION_COUNT=12
```

Deployment provisions or updates the Rayfin backend, applies the schema, builds the React application and publishes the Fabric App. The exact hosting origin must be registered as an SPA redirect URI.

11.3 First sync and upgrades

The first deployment shows a guided synchronization gate. The same gate appears when the major or minor snapshot contract changes. One sync may require several UDF invocations: an authoritative base scan followed by resumable deep-metadata slices. The progress display reports the active phase and completed items without inventing progress while a Fabric request is still running.

Only the configured synchronizer can publish the first snapshot or a later refresh. If another user reaches the gate, Atlas shows the configured account to contact.

After a successful refresh, current data and history switch together. The active view is remounted against the new snapshot so stale selections do not survive into an incompatible graph.

11.4 Routine operation

The expected operating pattern is simple:

1. Restrict the Fabric App audience to people allowed to read the full workspace metadata graph.
2. Keep the synchronizer account and UDF permissions current.
3. Run synchronization after material workspace changes or on the team's chosen review cadence.
4. Review Radar for new priority risks.
5. Use Change Center when the team needs exact before and after evidence.
6. Track low posture pillars through the linked catalog, access or operations view.
7. Rotate the trusted writer explicitly when the synchronization owner changes.

12 Coverage and known limits

Atlas always indexes top-level Fabric items. Deeper inventory depends on the item type, tenant settings and what the relevant Fabric or Power BI endpoint returns.

Fabric element	Current depth	Important boundary
Lakehouse	Item metadata, tables, columns, SQL endpoint context and lineage	Schema-enabled variants use OneLake listing and SQL endpoint metadata refresh where needed
Warehouse and SQL Database	Item metadata, scanner schema, optional read-only SQL catalog, access, lineage and jobs	SQL schema remains unknown when the optional token or endpoint is unavailable
SQL endpoint	Item identity, scanner metadata, read-only catalog where available and storage-to-model bridge	The generated endpoint is still represented separately from its parent store
Semantic Model	Tables, columns, measures, descriptions, hidden flags, measure DAX and object dependencies	Requires scanner schema and expression options; ambiguous references are discarded
Report	Item metadata, bound model, documented owner and pages	Visual and field bindings are not exposed by this flow

Fabric element	Current depth	Important boundary
Notebook	Item metadata, lineage, access and recent runs	Source code and cells are not read
Data Pipeline	Item metadata, lineage, access and recent runs	Activities, expressions and definitions are not copied
Dataflow and Datamart	Item metadata, documented owner and official upstream IDs	Query content is not copied and cross-workspace dependencies are omitted
Eventhouse and KQL Database	Item metadata, Kusto tables, columns, functions, materialized views, lineage, access and jobs	Business rows, query text and policy bodies are not copied
Ontology, Graph Model and Data Agent	Definition metadata, selected elements, relationships and object lineage	Definition access uses a separate optional write-scoped token
Fabric App, AppBackend and User Data Function	Top-level item and exposed relationships	Internal service inventory and function source are not copied

12.1 Version 1.x scope

Version 1.x indexes one configured workspace per deployment. It is not an automatic tenant-wide crawler.

Multi-workspace support is planned as a controlled scope. The roadmap first adds independent workspace manifests and refresh queues, then aggregates catalog experiences, and later adds verified cross-workspace lineage only when both endpoints are inside the indexed scope.

12.2 Reading unknown and not applicable states

Three states must remain separate:

- A value is present.
- The metadata family was collected and the value is empty.
- The metadata family was not exposed or was not applicable.

Atlas uses capability flags and explicit N/A states to preserve that difference. This prevents a missing API response from becoming a false governance failure.

13 Open source implementation

Fabric Atlas is MIT licensed. The deployed About page exposes the current version, build identifier, source repository, release history and clone command.

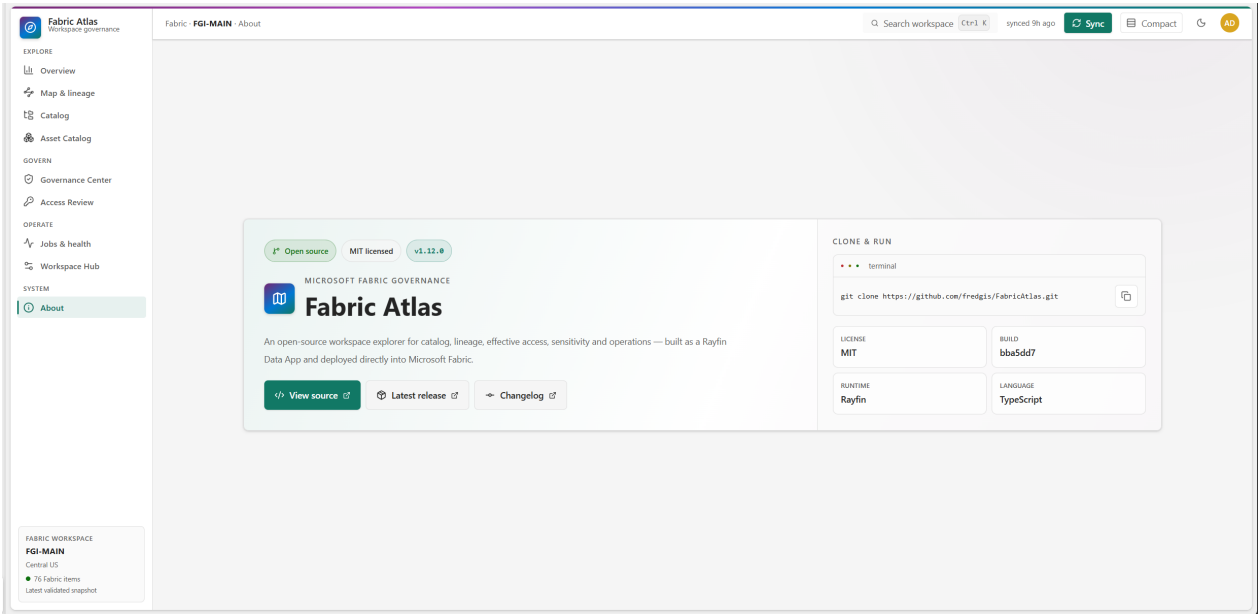


Figure 22: The About page for version 1.12.0 in the example deployment.

The front end uses React 19, TypeScript, Vite, Tailwind CSS and Radix primitives. Rayfin defines and provisions the authenticated data application. The metadata scan runs in a Fabric User Data Function.

Repository area	Responsibility
src/App.tsx	Application shell and hash-based navigation
src/atlas/model.ts	Shared UI data model
src/atlas/store.tsx	Hydration, synchronization and comments
src/atlas/live-sync.ts	User Data Function invocation and response mapping
src/atlas/backend.ts	Rayfin persistence and trusted snapshot loading
src/atlas/lineage.ts	Lineage normalization, traversal, impact and layout
src/atlas/schema-lineage.ts	DAX-resolved schema object dependencies
src/atlas/history.ts	Validated snapshot comparison and trends
src/atlas/views/	Product pages
rayfin/data/	Persisted entities and row policies
fabric/udf/atlas_sync_functions/	Server-side Fabric metadata collection

The project uses strict TypeScript checking and validates changes with:

```
npm test
npm run lint
npm run build
```

The production build is intentionally straightforward for the current application size. Version 1.12.0 ships one main application chunk of about 1.0 MB minified, or about 0.28 MB gzip, plus a smaller Rayfin dependency chunk. That tradeoff is documented rather than hidden behind an ineffective dynamic import.

14 Conclusion

Fabric Atlas gives a workspace team one consistent metadata snapshot and several ways to read it. The same indexed evidence supports catalog search, lineage, DAX dependency analysis, access review, governance findings, snapshot comparison and operations.

Its limits are as important as its features. Atlas does not store business rows, does not copy source code and does not claim report field usage that the current APIs do not expose. Within those boundaries, it provides a practical answer to three recurring questions: what exists, what changed and what could be affected.

15 Appendix A: figure index

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16 Appendix B: glossary

Term	Meaning in Fabric Atlas
Active snapshot	Newest complete trusted snapshot selected by its Workspace manifest
Capability	Evidence that a metadata family was collected or was unavailable
Consumer	Item or schema object that depends on the selected source
DAX edge	Dependency resolved from a measure expression to one synchronized object
Effective access	Highest permission produced by all workspace and item grants
Finding	Reproducible governance check evaluated from synchronized evidence
Inferred object hop	Cross-item source connection requiring item lineage and one unique schema match

Term	Meaning in Fabric Atlas
Radar occurrence	New priority finding or risky change in the latest adjacent snapshot pair
Reachable pair	Principal and item combination with effective access
Trusted writer	Configured immutable Entra subject allowed to publish and prune snapshots

17 References

- Fabric Atlas source: <https://github.com/fredgis/FabricAtlas>
- Fabric Atlas releases: <https://github.com/fredgis/FabricAtlas/releases>
- Rayfin source: <https://github.com/microsoft/rayfin>
- Microsoft Fabric documentation: <https://learn.microsoft.com/fabric/>